



**Ghulam Ishaq Khan Institute of Engineering Sciences  
and Technology (GIKI)**

**Project Scope Document**

**For**

**ARSS: Autonomous Response System for SOC**

Version 1.0

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# Table of Contents

## Contents

|                                                            |   |
|------------------------------------------------------------|---|
| Abstract .....                                             | 3 |
| 1. Introduction .....                                      | 4 |
| 2. Problem Statement .....                                 | 4 |
| 3. Problem Solution/Objectives of the Proposed System..... | 5 |
| 3.1 Objectives.....                                        | 5 |
| 4. Related System Analysis/Literature Review .....         | 6 |
| 5. Vision Statement .....                                  | 7 |
| 6. Scope .....                                             | 7 |
| 7. Project Stakeholders and Roles .....                    | 8 |
| 8. References .....                                        | 8 |

## Abstract

Security Operations Centers (SOCs) receive thousands of alerts daily from network monitoring tools, yet the majority are false positives or low-priority events that still consume analyst time. This volume creates alert fatigue, a condition where analysts become desensitized and begin missing genuine threats, not because detection failed but because meaningful review at scale is not humanly possible. Existing automated tools, including rule-based Security Orchestration, Automation and Response (SOAR) platforms and machine learning scoring systems, address detection and ranking but do not take response actions and do not learn from outcomes. This project proposes ARSS, an Autonomous Response System for SOC, which sits at the Security Information and Event Management (SIEM) output layer and makes autonomous triage decisions using a Reinforcement Learning (RL) agent trained to choose one of four actions: Ignore, Log, Block, or Isolate. The agent's reward function is grounded in the MITRE Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK) framework, ensuring response decisions reflect real-world threat severity rather than abstract training costs. To build analyst trust, every decision is accompanied by a plain-English explanation generated by a Large Language Model (LLM), drawing on feature-level evidence from SHapley Additive exPlanations (SHAP) and the RL decision rationale, making every automated action auditable and verifiable by the analyst.

# 1. Introduction

Security Operations Centers handle thousands of alerts daily from Intrusion Detection Systems (IDS), firewalls, and SIEM platforms. Detection tools have matured significantly: suspicious activity is flagged at scale and aggregated reliably. The problem that remains unsolved is triage, deciding what to actually do with each flagged alert before it reaches the analyst.

Existing Security Orchestration, Automation and Response (SOAR) tools address this through rule-based playbooks. These work well for known, anticipated attack patterns, but break down on new attack types and require manual updates every time the threat landscape shifts. Critically, they do not learn. When an analyst overrides a playbook decision, that correction is discarded. The system makes the same type of error again.

ARSS sits at the SIEM output layer and replaces static playbooks with a trained RL policy. The policy takes the attack type, detection confidence, and danger score as input, and outputs a triage action. Because the reward function is grounded in the MITRE ATT&CK framework, the agent learns that the cost of missing a Credential Access attack is far higher than missing a Reconnaissance scan. Every action is explained to the analyst in plain English through an LLM narrative, making the system auditable and trustworthy in a real SOC environment.

# 2. Problem Statement

SOC analysts routinely handle more alerts than they can meaningfully review. The vast majority are either benign or low-priority, but no reliable automated system exists to separate and handle them without analyst involvement. Current rule-based SOAR tools are static and cannot adapt. Machine learning scoring systems rank alerts but do not take response actions. Neither approach learns from analyst feedback over time.

The deeper issue is that existing automated triage tools, including academic RL systems, optimize for abstract performance metrics disconnected from how security teams actually assess threats. A system whose decisions analysts cannot understand or verify will not be trusted or adopted, regardless of its accuracy numbers. There is a need for an adaptive triage agent that grounds its decisions in established threat intelligence, takes concrete response actions, and explains every decision in terms the analyst already understands.

### 3. Problem Solution/Objectives of the Proposed System

ARSS addresses this through three components. A two-stage detection ensemble (XGBoost + Deep Neural Network (DNN)) classifies incoming network flows as benign or attack, then categorizes confirmed attacks into seven types with a confidence score. These stages serve as training infrastructure; in deployment, the SIEM provides equivalent data directly.

A Deep Learning triage agent receives a state vector of danger score, attack category, and classification confidence, and outputs one of four response actions. The reward function uses MITRE ATT&CK severity weights, so missed high-severity attacks are penalized proportionally to their real-world impact. A Cognitive Semantic Layer then combines SHAP feature attributions from the detector with the RL decision and passes both to an LLM, which generates a short plain-English incident narrative displayed on the SOC dashboard.

#### 3.1 Objectives

**BO-1:** *To build a two-stage ML ensemble that classifies network flows and produces attack category labels with confidence scores for RL state construction.*

**BO-2:** *To design a category-conditioned RL state space encoding danger score, attack category, and classification confidence, enabling category-specific response policies.*

**BO-3:** *To train a Deep Learning agent with a MITRE ATT&CK severity-weighted reward function that optimizes the tradeoff between analyst workload reduction and threat containment.*

**BO-4:** *To implement a Cognitive Semantic Layer using an LLM to generate plain-English incident narratives from SHAP attributions and RL decisions for the SOC dashboard.*

**BO-5:** *To evaluate the system on the CIC-IIoT 2025 dataset and demonstrate measurable reduction in analyst-escalated alerts without proportional loss in threat containment.*

**BO-6:** *To compare the RL agent against a rule-based threshold baseline to establish the contribution of the learned policy.*

## 4. Related System Analysis/Literature Review

**Table 1: Related System Analysis with Proposed Project Solution**

| Application Name                                                   | Features                                                                                            | Weakness                                                                              | Relevance with Proposed System                                                                        |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| SAC-AP -- Chavali et al., IEEE CEC 2022 [2]                        | Soft Actor-Critic RL for alert prioritization. ~16% gain over DDPG.                                 | Synthetic game-theoretic reward. Ranking output only, no response actions.            | ARSS uses MITRE ATT&CK reward and outputs discrete response actions.                                  |
| TD3-AP -- Chavali et al., Computers & Security 2024 [3]            | Extends SAC-AP with TD3. Validated on real IDS datasets.                                            | Same abstract reward. No CTI grounding, no response actions.                          | ARSS adopts off-policy RL but grounds reward in MITRE ATT&CK with 4-action output.                    |
| KNAP -- Chavali et al., AINA 2024 [4]                              | Attack graph knowledge injection into DRL. Better sample efficiency.                                | Custom heuristic knowledge, not an established threat framework. No response actions. | ARSS uses MITRE ATT&CK as the knowledge source instead of a custom heuristic.                         |
| AlertPro -- Wang et al., Computers & Security 2024 [5]             | RL with analyst feedback for alert re-ranking. Sub-500ms latency.                                   | Binary feedback signal, no severity weighting. Ranking only, no response actions.     | ARSS adds MITRE severity-weighted feedback and a discrete 4-action response space.                    |
| RADAMS -- Huang & Zhu, Computers & Security 2022 [6]               | RL alert de-emphasis against flooding. 95.89% recall on 500K+ alerts.                               | Attention management only. No response actions, no CTI grounding.                     | ARSS takes actual response actions. LOG covers deferral; Block and Isolate cover autonomous response. |
| Alert Fatigue Survey - - Tariq et al., ACM CSUR 2025 [1]           | Systematic review. 4,484 alerts/day, 67% uninvestigated. Flags RL at triage layer as underexplored. | Survey only, no implementation.                                                       | Establishes the problem ARSS solves and confirms the gap is peer-recognized.                          |
| Alert Prioritization Survey -- Jalalvand et al., ACM CSUR 2024 [7] | Taxonomy of prioritization methods. Flags CTI-grounded RL as least-explored.                        | Survey only, no implementation.                                                       | Confirms ARSS sits in an identified, unaddressed gap in the solution space.                           |
| A2C Framework -- Chhetri et al., ACM TOIT 2024 [8]                 | Automate-Augment-Collaborate framework for SOC automation modes.                                    | Conceptual only, no RL policy.                                                        | ARSS operationalizes A2C: LOG = Collaborate, remaining actions = Automate.                            |
| MITREtrieval -- Huang et al., IEEE TNSM 2024 [9]                   | BERT + MITRE ATT&CK for TTP extraction from threat reports.                                         | Does not feed TTPs back into a triage policy or reward function.                      | Validates MITRE ATT&CK as ML-compatible. ARSS uses it directly in the RL reward.                      |
| CrowdStrike Charlotte AI -- Commercial, 2024 [10]                  | Imitation learning on analyst decisions. LLM explanation layer.                                     | Cannot exceed average analyst performance. Replicates historical bias.                | ARSS optimizes beyond past decisions using RL with a MITRE-grounded reward.                           |

## 5. Vision Statement

ARSS is an autonomous triage agent designed to protect analyst attention, not replace it. It handles routine alerts automatically, escalates genuine threats with full explanations, and grounds every decision in the MITRE ATT&CK framework so that security teams can trust and audit what the system does. Gartner's 2026 cybersecurity trends report identifies alert triage automation as a key near-term improvement for security operations teams, while stressing that human-in-the-loop frameworks remain critical as these systems mature [11]. ARSS is built for exactly that balance: autonomous action on routine cases, full transparency on everything else.

## 6. Scope

ARSS covers the design, development, and evaluation of an autonomous alert triage system with an explainability layer. The system will:

- Accept alert data from a SIEM or simulated alert stream as input.
- Classify alerts through a two-stage ML ensemble into binary and categorical threat labels.
- Generate a state vector of danger score, attack category, and confidence for the RL agent.
- Output one of four triage actions (Ignore, Log, Block, Isolate) using a trained Deep Learning agent policy.
- Produce SHAP feature attributions for each classification decision.
- Generate a plain-English incident narrative using an LLM from SHAP output and RL decision.
- Display decisions and explanations on a SOC analyst dashboard.

The project will focus on prototype-level implementation evaluated on the CIC-IIoT 2025 dataset with simulated alert replay. Real-time packet capture, live SIEM integration, and production deployment are outside the current scope. Constraints include dataset availability, computational resources, and timeline. The system is designed to be modular so that live SIEM integration can be added in a later phase without architectural changes.

## 7. Project Stakeholders and Roles

Table 2: Project Stakeholders for ARSS

| Stakeholder                                   | Role                                                                  |
|-----------------------------------------------|-----------------------------------------------------------------------|
| <b>GIKI</b>                                   | Project sponsor                                                       |
| <b>Muhammad Daniyal (2023406)</b>             | System design, Literature review, RL agent development, documentation |
| <b>Haider Iqbal (2023416)</b>                 | ML ensemble training, Security, testing                               |
| <b>Syed Daud (2023677)</b>                    | Explainability layer, dataset preprocessing, dashboard development    |
| <b>Dr Muhammad Fawad Khan</b>                 | Technical guidance, research direction, academic oversight            |
| <b>Miss Hadia Abbas</b>                       | Technical guidance, security principles                               |
| <b>Senior Design Project Evaluation Panel</b> | Project assessment and final evaluation                               |

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